

MODEL-AGNOSTIC – PDP

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Introduction

This project aims to apply model-agnostic interpretability techniques, specifically Partial Dependence Plots (PDPs), to analyze the behavior of Random Forest-based regression models. Throughout the exercise, a practical and systematic approach was followed, covering all key stages of the predictive analysis and interpretability process:

First, real-world data related to bicycle rentals and housing prices were loaded, cleaned, and sampled. Second, Random Forest regression models were trained, fine-tuned to predict the variable of interest (cnt in the case of bicycles, and price in the case of housing). In addition, one- and two-dimensional PDPs were generated, showing how different explanatory variables influence the model's prediction. Finally, a detailed analysis of the results was performed, drawing conclusions about the model's behaviour and the relationship between the variables and the target variable. In the first block of the project, a model is trained to estimate the number of daily bicycle rentals based on time, temperature, humidity, and wind, visualizing both the individual influence of each variable and the interaction between temperature and humidity. In the second block, a model is trained on a real estate dataset to predict home prices based on characteristics such as the number of bedrooms, bathrooms, square footage, and number of floors.

The entire process was developed in R, using specific libraries such as randomForest, pdp, ggplot2, and dplyr, and follows reproducible logic that allows black-box models to be interpreted from a comprehensible and visual perspective.

1. PDP unidimensional – Bike Rentals

In this first part of the project, one-dimensional Partial Dependence Plots (PDPs) were used to analyse the influence of several explanatory variables on the predicted number of rented bicycles (cnt). Before training the model, the bike rental dataset was preprocessed by converting the date variable into a numerical variable called days_since_2011, representing the number of days since the beginning of 2011. Missing values were checked and removed if necessary, and only the variables required for the analysis were selected.

A Random Forest regression model was then trained using days_since_2011, temp, hum and windspeed as predictor variables. The model achieved a good predictive performance, explaining approximately 86.81% of the variance of the target variable. In addition, the variable importance analysis showed that days_since_2011 and temperature were the most influential variables in the prediction of bike rentals.

To better understand the behaviour learned by the model, one-dimensional PDPs were generated for each explanatory variable. These plots allow the marginal effect of each variable on the predicted number of bike rentals to be visualised while averaging the remaining variables.

The analysis for each variable is presented below:

1.1. Instant (días desde 2011)

The graph shows a slight upward trend in the number of bicycles rented over the course of the day since 2011. This trend suggests a positive trend in the use of the rental service, possibly linked to greater implementation of the system, improvements in cycling infrastructure, or a change in urban mobility habits.

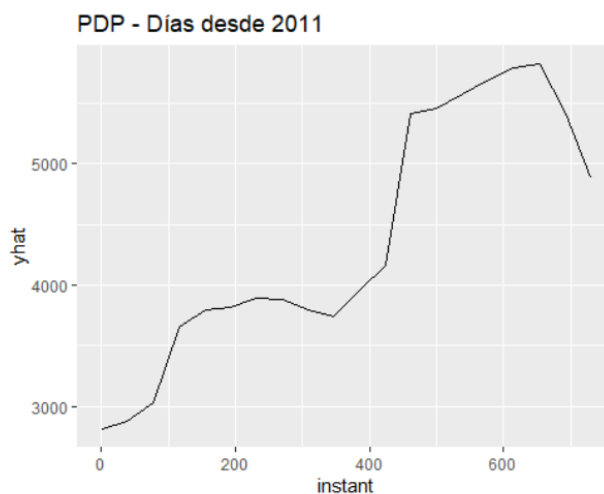


Figure 1 → PDP - Days since 2011

1.2. Temp (temperatura)

In this case, the graph shows a clearly positive relationship between temperature and the number of rentals. As temperature increases, so does the number of bicycles rented. The increase is not linear: the sharpest increase is observed with average temperature values. This result is consistent with the expected behavior, as good weather generally favors the use of bicycles as a means of transportation or leisure activity.

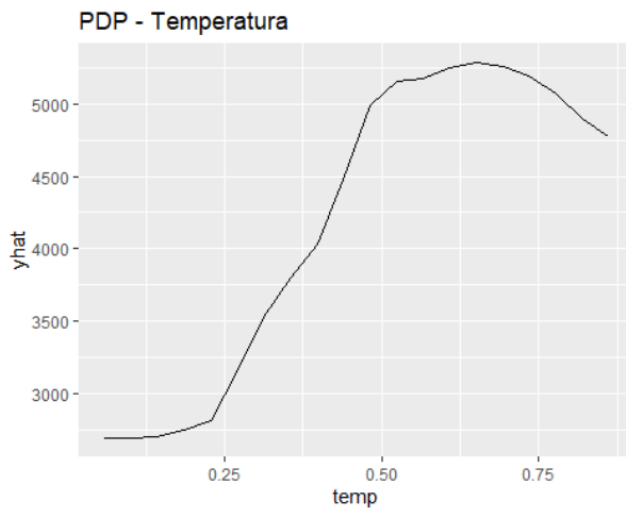


Figure 2 → PDP – Temperature

1.3. Hum (humedad)

The relationship between humidity and the number of rentals is inverse. As relative humidity increases, the model's prediction indicates a progressive decrease in the number of rented bicycles. This can be interpreted as a preference among users to avoid traveling in high humidity conditions, which are often associated with discomfort or bad weather.

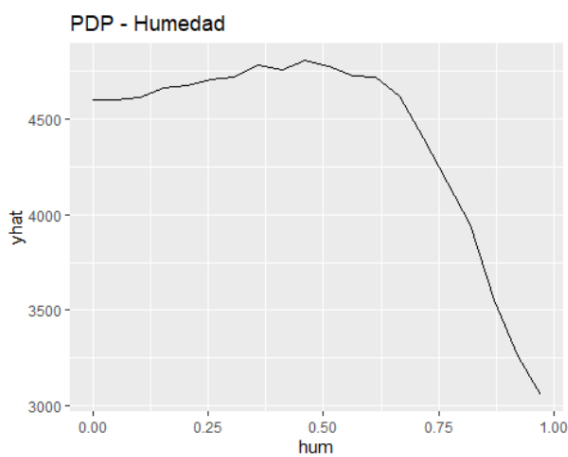


Figure 3 → PDP – Humidity

1.4. Windspeed (velocidad del viento)

Finally, the graph for wind speed also shows a negative relationship. In high wind conditions, the number of rentals decreases. This pattern makes sense from a practical standpoint, as cycling in strong winds is more difficult and less comfortable, which can discourage users.

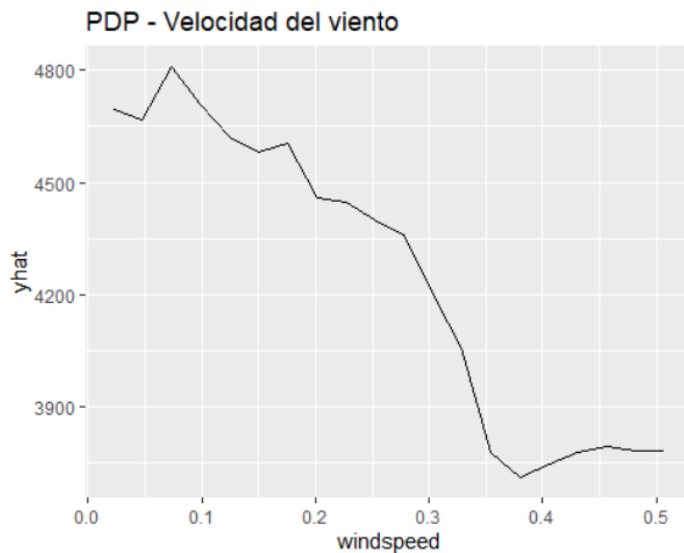


Figure 4 → PDP - Wind speed

Taken together, these graphs clearly identify the variables that most influence the model's prediction. According to the importance calculated during Random Forest training, the variable instant is the most influential, followed by temp, hum, and finally windspeed. However, from an interpretive perspective, temp stands out as one of the variables most clearly associated with an increase in rents, while hum and windspeed are related to reductions.

2. PDP bidimensional – Bike Rentals

In this second section, the joint interaction of two explanatory variables on the prediction of the number of bicycles rented (cnt) is explored using a two-dimensional partial dependence plot (2D PDP).

Specifically, a PDP plot was generated using the variables temp (temperature) and hum (humidity), which showed significant effects individually in the previous section. The objective is to analyze how the combination of both variables affects the output of the Random Forest model, identifying joint patterns that are not apparent in one-dimensional PDPs.

Given the size of the original dataset, random sampling was applied to reduce the computational burden without compromising the quality of the analysis. The two-dimensional PDP was then generated using the `partial()` function of the `pdp` package and visually represented as a heat map with `geom_tile()`, adjusting the width and height of each cell to avoid gaps.

The resulting heat map represents the normalized temperature values (temp) on the X axis and the humidity values (hum) on the Y axis. The color indicates the number of bicycles rented predicted by the model (yhat), where more intense tones correspond to a higher number of rentals.

The results show that optimal conditions for a high number of rentals occur when the temperature is high (above average) and humidity is moderate or low. However, in scenarios where the temperature is low or humidity is very high, the rental prediction decreases significantly.

This behaviour is consistent with common sense: pleasant temperatures and a dry environment encourage bicycle use, while cold temperatures or high humidity levels can act as deterrents.

Furthermore, the analysis confirms that temperature has a more dominant effect than humidity. Although both factors play a role, differences in the number of predicted rentals are due more to changes in temperature than to variations in humidity.

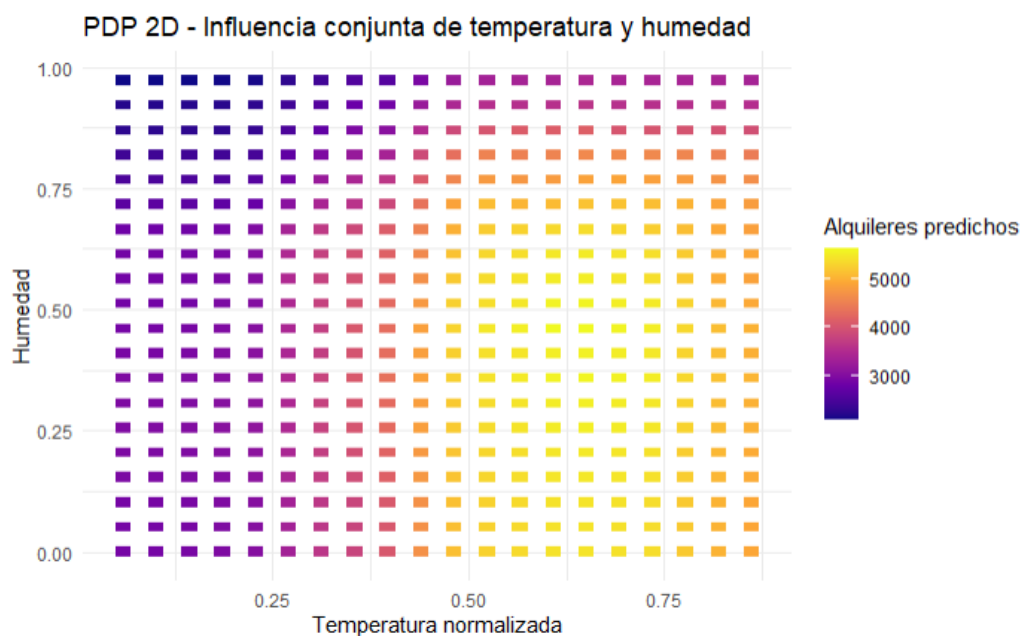


Figure 5 → PDP 2D - Joint influence of temperature and humidity

To complement the interpretation, density distributions for temp and humidity were generated on a random sample of the dataset. These graphs allow us to visualize the frequency with which certain temperature and humidity values appear in the data, which helps contextualize the information shown in the heat map.

The density of temperature shows a greater concentration in average values, while humidity has a more dispersed distribution, with a slight bias toward high values. This implies that although ideal rental conditions occur with low humidity, they are

not always the most frequent in the data.

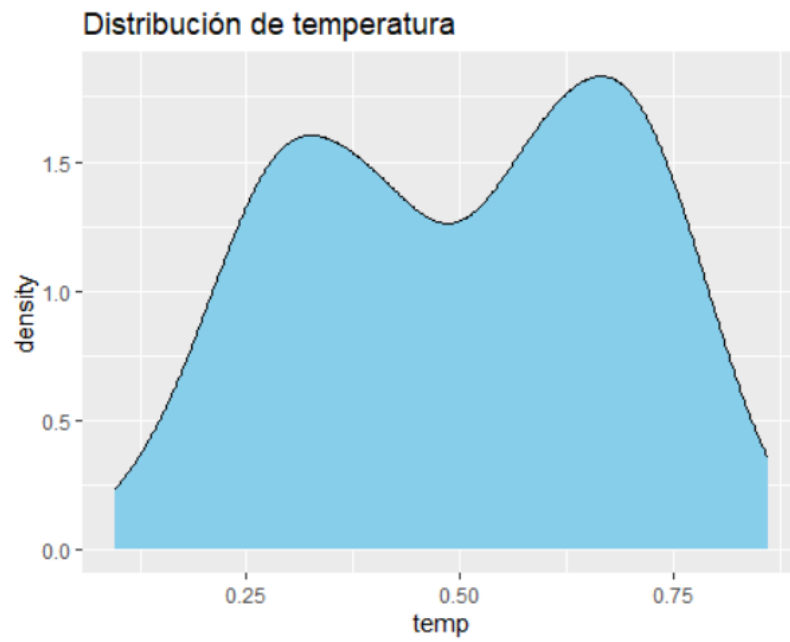


Figure 6 → Density distribution of temperature values

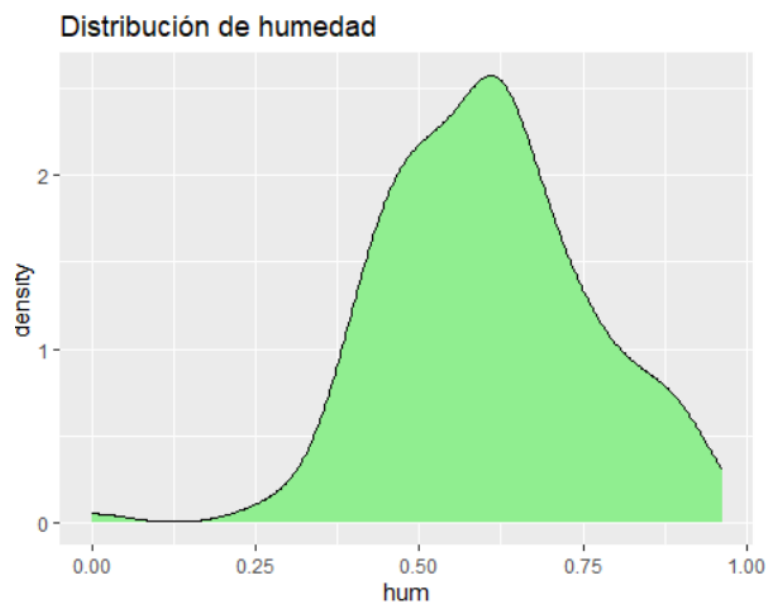


Figure 7 → Density distribution of humidity values

3. PDP unidimensional – Precio de vivienda

In this final part of the project, one-dimensional Partial Dependence Plots (PDPs) were applied to analyse the behaviour of a Random Forest regression model trained to predict house prices. The objective of this section is to understand how different structural characteristics of a property influence the value predicted by the model.

Before training the model, the `kc_house_data.csv` dataset was preprocessed. The date variable was converted into Date format, missing values were checked and removed if necessary, and only the variables required for the analysis were selected. In addition, unrealistic values were filtered out to ensure data consistency. Due to the size of the original dataset, a random sample of 300 observations was extracted in order to reduce computational cost while maintaining representative information for the PDP analysis.

The Random Forest model was trained using the variables bedrooms, bathrooms, sqft_living, sqft_lot, floors and yr_built as predictors of house price. According to the variable importance analysis, sqft_living was identified as the most influential variable in the prediction process, followed by bathrooms.

To interpret the behaviour learned by the model, one-dimensional PDPs were generated for the variables bedrooms, bathrooms, sqft_living and floors. These plots make it possible to visualise the marginal effect of each variable on the predicted house price while averaging the remaining explanatory variables.

3.1. Bedrooms (número de habitaciones)

The graph shows that the effect of this variable on the predicted price is nonlinear. Up to approximately 4 or 5 bedrooms, the price increases slightly. However, beyond that point, the relationship stabilizes and even decreases for some values. This indicates that a greater number of bedrooms does not necessarily imply a higher price, possibly because many homes with many bedrooms correspond to less desirable market segments or have limitations in other aspects (location, quality, etc.).

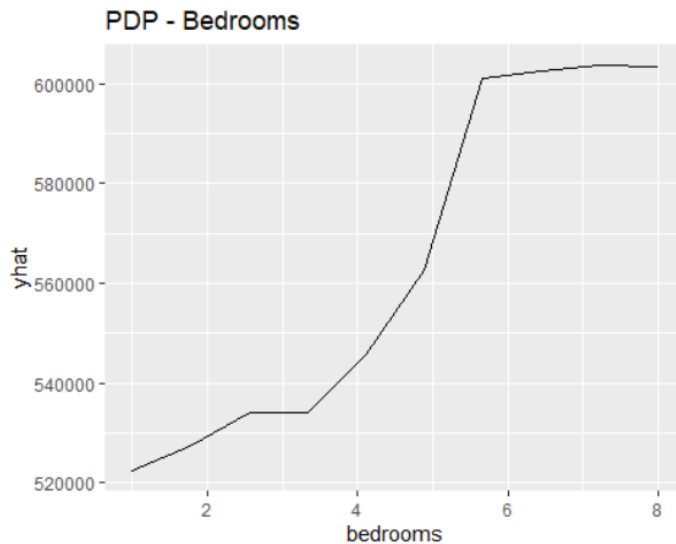


Figure 8 → PDP - Bedrooms

3.2. Bathrooms (número de baños)

The number of bathrooms is positively related to price. As this variable increases, the model predicts higher prices. However, the effect also levels off above a certain threshold (around 3 or 3.5 bathrooms). This result is reasonable, as having more than one bathroom is valued positively by buyers, although beyond a certain number it does not provide significant additional value.

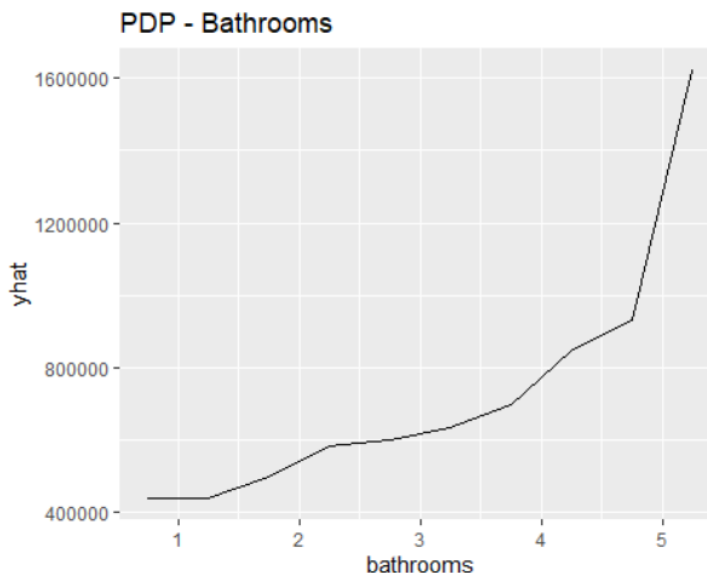


Figure 9 → PDP - Bathrooms

3.3. Sqft_living (superficie útil)

This is, without a doubt, one of the variables with the greatest impact on the predicted price. The graph shows a strongly increasing relationship: the larger the living area, the higher the price estimated by the model. The increase is almost

linear, especially in the mid-range, reflecting that available interior space is a fundamental determinant of a home's value.

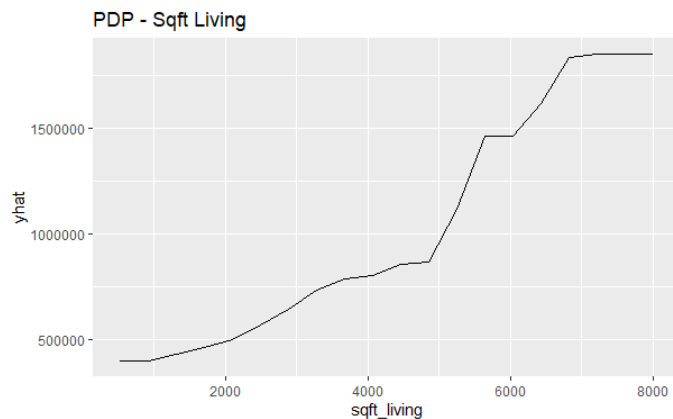


Figure 10 → PDP - Sqft Living

3.4. Floors (número de plantas)

In this case, the graph reveals that two-story homes tend to have higher values than single-story homes. However, the effect of the number of floors is not as pronounced as in the case of floor area. It is also observed that having more than two floors does not necessarily translate into higher prices, suggesting that this factor contributes to some degree of value.

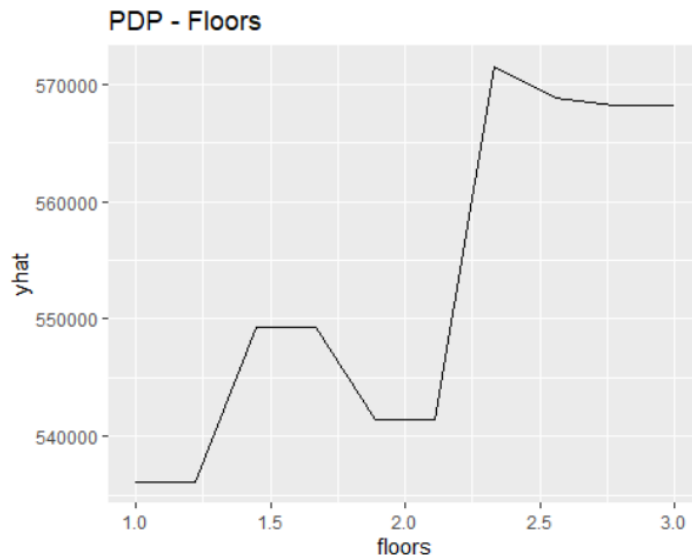


Figure 11 → PDP – Floors

Taken together, these results allow us to identify which structural characteristics have the greatest influence on the prediction of house prices. Among all the analysed variables, `sqft_living` clearly stands out as the most influential predictor, followed by bathrooms. In contrast, the effect of bedrooms and floors is more limited and presents nonlinear patterns. These findings are coherent with the idea

that usable living space is one of the main determinants of a property's market value.

Overall, the PDP analysis provides useful insights into the behaviour learned by the Random Forest model and helps explain how different housing characteristics contribute to the predicted price. This type of interpretability is especially valuable in real estate contexts, where understanding the factors that drive price estimation can support more informed decision-making.

Conclusion

Throughout this project, the usefulness of Partial Dependence Plots (PDPs) has been demonstrated as an effective model-agnostic interpretability technique for understanding the behaviour of complex predictive models, specifically Random Forest regression models.

By applying PDPs to two real-world problems — bike rental prediction and house price estimation — it has been possible to visually and intuitively analyse the relationship between explanatory variables and model predictions. The one-dimensional PDPs revealed important patterns, such as the positive effect of temperature on bike rentals and the strong influence of living area on housing prices. In addition, the two-dimensional PDP provided a deeper understanding of the joint interaction between temperature and humidity in the prediction of bike rental demand.

The preprocessing stage also played an important role in ensuring data quality and improving the interpretability of the analysis. Variables were properly selected and transformed before training the models, allowing the PDPs to represent meaningful relationships between predictors and target variables.

More generally, this project highlights the importance of interpretability techniques in modern machine learning. Although Random Forest models can behave as black-box algorithms, PDPs make it possible to translate complex model behaviour into visual and understandable patterns. In contexts where transparency, explainability and trust are increasingly important, these techniques become essential tools for supporting data-driven decision-making.